

GENESIS AI ENGINEERING HANDBOOK

SPRINT 1B – DELIVERABLE EF-02

FUNCTIONAL SPECIFICATION DOCUMENT (FSD)

Project: Genesis AI

Product: Genesis Product Factory

Version: 1.0

Status: Draft

1. Purpose

This document defines the functional behavior of Genesis Product Factory.

Unlike the Business Requirements Specification (BRS), which explains *why* Genesis exists, the Functional Specification defines *how the product behaves* from a functional perspective.

No implementation shall be performed unless a corresponding functional requirement exists.

2. Functional Scope

Genesis Product Factory shall provide the following major capabilities:

FR-100 Research

FR-200 Decision Engine

FR-300 Product Planning

FR-400 Creative Generation

FR-500 Marketing

FR-600 Publishing

FR-700 Analytics

FR-800 Knowledge Management

FR-900 Administration

3. Functional Module – Research

FR-101

Accept Business Request

Description

Genesis shall accept a business idea or niche from the Founder.

Input

Business Idea

Output

Project Record

Acceptance Criteria

Project successfully created.

FR-102

Validate Input

Genesis validates:

Business Name

Niche

Language

Required Information

Missing information results in a clarification request.

FR-103

Trend Analysis

Genesis shall:

Research market trends

Identify growth

Identify decline

Identify seasonality

Generate Trend Score

FR-104

Competitor Analysis

Genesis shall:

Identify competitors

Estimate competition

Find market gaps

Recommend differentiation

FR-105

Customer Research

Genesis shall generate:

Customer Personas

Pain Points

Buying Motivation

Purchase Barriers

Expected Lifetime Value

FR-106

Product Research

Genesis shall recommend:

Products

Estimated Margin

Demand

Risk

Difficulty

FR-107

Research Report

Genesis compiles:

Trend Report

Competitor Report

Customer Report

Product Report

↓

Research Report

4. Functional Module – Decision Engine

FR-201

Receive Research Report

CEO receives completed report.

FR-202

Evaluate Business

Genesis calculates:

Demand

Competition

Profit

Brand Potential

AI Compatibility

Risk

Overall Score

FR-203

Business Decision

Possible results:

Approve

Reject

Needs Investigation

FR-204

Store Decision

Decision stored permanently.

5. Functional Module – Product Planning

FR-301

Create Product Blueprint

Inputs:

Research

Decision

Brand

Outputs:

Blueprint

FR-302

Recommend Product Categories

Examples

T-Shirts

Posters

Mugs

Canvas

Hoodies

Tote Bags

FR-303

Collection Planning

Genesis groups products into:

Collections

Bundles

Launch Waves

FR-304

Pricing Strategy

Generate:

Cost

Margin

Retail Price

Suggested Discounts

6. Functional Module – Creative

FR-401

Generate Creative Brief

FR-402

Generate Design Concepts

FR-403

Generate AI Prompts

FR-404

Generate Mockup Prompts

FR-405

Brand Consistency Validation

7. Functional Module – Marketing

FR-501

Generate SEO

FR-502

Generate Product Title

FR-503

Generate Description

FR-504

Generate Instagram Content

FR-505

Generate Facebook Content

FR-506

Generate Marketing Pack

8. Functional Module – Publishing

FR-601

Generate Publishing Checklist

FR-602

Validate Marketplace Compatibility

FR-603

Generate Publishing Package

9. Functional Module – Analytics

FR-701

Track Product Performance

FR-702

Track Sales

FR-703

Track Marketing Performance

FR-704

Generate Recommendations

10. Functional Module – Knowledge

FR-801

Store Reports

FR-802

Store Prompts

FR-803

Store Templates

FR-804

Store Lessons Learned

11. Functional Module – Administration

FR-901

Manage Employees

FR-902

Manage Departments

FR-903

Manage Versions

FR-904

Manage Architecture

12. Use Cases

UC-001

Founder creates a new business.

Actor

Founder

Flow

Input Idea

↓

Research

↓

CEO Review

↓

Product Plan

↓

Creative

↓

Marketing

↓

Publishing Package

↓

Founder Approval

UC-002

Research Department executes.

Actor

Research Department

Flow

Receive Project

↓

Research

↓

Generate Reports

↓

Submit CEO

UC-003

Creative Department executes.

Flow

Receive Blueprint

↓

Generate Assets

↓

Submit Marketing

13. Functional Rules

FRR-001

Every project begins with research.

FRR-002

No department executes without approved input.

FRR-003

Departments cannot skip workflow stages.

FRR-004

Every output is versioned.

FRR-005

Every workflow generates reports.

FRR-006

Every report is permanently stored.

14. Validation Rules

Project Name required.

Business Type required.

Research complete before CEO review.

Approved Blueprint required before Creative.

Creative Pack required before Marketing.

Marketing Pack required before Publishing.

Founder Approval required before Release.

15. Error Handling

Possible Errors

Invalid Input

Missing Information

Research Failure

Approval Failure

Publishing Failure

Workflow Timeout

Each error generates:

Error Code

Description

Recovery Action

Audit Log

16. Permissions

Founder

Full Access

CEO AI

Decision Authority

Department Heads

Department Management

Employees

Execution Only

17. Traceability Matrix

Every Functional Requirement maps to:

Business Requirement

↓

Department

↓

Employee

↓

Database Table

↓

API

↓

Sprint

This ensures every feature has implementation ownership.

18. Future Functional Expansion

Future modules include:

CRM

Inventory

Finance

Supplier Management

Legal

Manufacturing

Customer Support

Each future function will follow the same numbering convention.

CTO Review

Approved with Recommendations

The FSD now establishes **what Genesis must do**, but before implementation I want one more refinement in EF-03:

- Every functional requirement should be accompanied by **sequence diagrams**.
- Every module should define **input and output JSON schemas**.
- Every workflow should include **state transition diagrams**.
- Every requirement should specify **performance expectations** (e.g., response time, retry behavior).

These additions will make the Functional Specification directly implementable by developers and AI coding agents.

Document Version: 1.0

Status: Draft

Founder Approval: Pending

Sprint 1B Progress: 2 / 8 Engineering Deliverables Complete